

Rajneil Baruah

Particle Physicist | BSM Phenomenology | Machine Learning & Statistical Inference

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Education

Ph.D. in Physics, Bennett University

2022–Ongoing

- Thesis: *Deciphering Anomalous Results in Particle Physics using Statistical and Machine Learning Tools*
- Advisor: [Dr. Subhadeep Mondal](#)

M.Sc. in Physics, University of Delhi

2018–2020

- Dissertation: *Solitary Waves and Solitons: Some Non-Perturbative Semi-Classical Methods in Quantum Field Theory*
- Advisor: [Prof. Debajyoti Choudhury](#)

B.Sc. (Honours) in Physics, University of Delhi

2015–2018

- Specialization: Theoretical Physics

Research Experience

Junior Research Fellow (Visitor), Bangabasi Evening College, University of Calcutta

2024–Present

- Project: A “Deep” Analysis Framework for Physics Beyond Standard Model (CRG/2022/003208)
- Principal Investigator: [Dr. Sunando Kumar Patra](#); Co-Principal Investigator: [Dr. Subhadeep Mondal](#)

Research Profile

My research lies at the interface of particle-physics phenomenology, statistical inference, and machine learning. I develop computational methods for exploring and constraining Beyond the Standard Model (BSM) parameter spaces and for extracting new-physics signatures from collider data, with particular emphasis on efficient posterior inference, jet tagging, and scalable ML architectures.

Research Interests

- **BSM phenomenology:** collider studies of non-minimal Higgs sectors (2HDMs and triplet extensions), seesaw models, RPV-MSSM, and 3-3-1 models, with emphasis on constraining model parameter spaces using experimental data.
- **Machine learning for collider physics:** ML-driven jet-tagging methods for identifying and probing new-physics signatures, including computationally efficient and lightweight architectures.
- **Statistical and generative inference:** posterior inference using normalizing flows and related ML methods, including approaches that accelerate traditional MCMC and nested-sampling workflows.

Research Skills

- **Particle phenomenology:** SARAH, SPheno, MadGraph, Pythia, Delphes, HiggsTools, OptEx
- **Statistical inference:** Bayesian inference, MCMC, nested sampling, normalizing flows, conditional flows, posterior estimation
- **Machine learning:** PyTorch, TensorFlow, Mathematica (MXNet); MLPs, autoencoders, VAEs, CNNs, GNNs, attention mechanisms, transformers
- **Scientific computing and data analysis:** Python, C/C++, Fortran, Mathematica; ROOT, NumPy, pandas, matplotlib

Publications

Journal Articles

- [1] R. Baruah, S. Mondal, S. K. Patra, and S. Roy, "Normalizing flow-assisted nested sampling on Type-II Seesaw model," *Eur. Phys. J. C*, vol. 85, no. 7, p. 816, Jul. 1, 2025. DOI: [10.1140/epjc/s10052-025-14502-5](https://doi.org/10.1140/epjc/s10052-025-14502-5) arXiv: [2501.16432](https://arxiv.org/abs/2501.16432) [hep-ph].
- [2] R. Baruah, A. Choudhury, K. Ghosh, S. Mondal, and R. Sahu, "Probing sub-TeV Higgsinos aided by a machine-learning-based top tagger in the context of trilinear R-parity violating SUSY," *Phys. Rev. D*, vol. 111, no. 9, p. 095004, May 1, 2025. DOI: [10.1103/PhysRevD.111.095004](https://doi.org/10.1103/PhysRevD.111.095004) arXiv: [2412.11862](https://arxiv.org/abs/2412.11862) [hep-ph].
- [3] R. Baruah, S. Mondal, S. K. Patra, and S. Roy, "Probing intractable beyond-standard-model parameter spaces armed with machine learning," *Eur. Phys. J. ST*, vol. 233, no. 15-16, pp. 2597–2618, Jul. 1, 2024. DOI: [10.1140/epjs/s11734-024-01236-w](https://doi.org/10.1140/epjs/s11734-024-01236-w) arXiv: [2404.02698](https://arxiv.org/abs/2404.02698) [hep-ph].

Preprints

- [4] R. Baruah, "Amortized inference of multi-modal posteriors using likelihood-weighted normalizing flows," 2025. arXiv: [2512.04954](https://arxiv.org/abs/2512.04954) [cs.LG].
- [5] R. Baruah, S. Mondal, S. K. Patra, and S. Roy, "Cnn on 'top': In search of scalable & lightweight image-based jet taggers," 2025. arXiv: [2512.05031](https://arxiv.org/abs/2512.05031) [hep-ph].

Manuscript in Preparation

- [6] R. Baruah, S. Mondal, S. K. Patra, and S. Roy, "This is sparta: A modular, efficient and scalable transformer architecture for jet-tagging," 2026, Manuscript in preparation.

Conference Proceedings

- [7] R. Baruah, S. Mondal, S. K. Patra, and S. Roy, "BSM Parameter Space Exploration with ML-Assisted Nested Sampling," *Springer Proc. Phys.*, vol. 432, pp. 409–412, 2026. DOI: [10.1007/978-981-95-1513-4_93](https://doi.org/10.1007/978-981-95-1513-4_93)
- [8] R. Baruah, S. Mondal, S. K. Patra, and S. Roy, "Boosting Beyond Standard Model Searches: ML-Enhanced Nested-Sampling for Rapid Parameter Estimation," *Springer Proc. Phys.*, vol. 432, pp. 445–448, 2026. DOI: [10.1007/978-981-95-1513-4_102](https://doi.org/10.1007/978-981-95-1513-4_102)
- [9] R. Baruah, A. Choudhury, K. Ghosh, S. Mondal, and R. Sahu, "Probing Sub-teV Higgsinos in a Trilinear RPV SUSY Scenario Using GNN-Based Boosted Top Tagger," *Springer Proc. Phys.*, vol. 432, pp. 357–360, 2026. DOI: [10.1007/978-981-95-1513-4_81](https://doi.org/10.1007/978-981-95-1513-4_81)

Selected Talks and Seminars

- "The Shape of Physics: Bayesian Inference, Jet Geometry, and the Architectures for Discovery", HEP-Theory Division Seminar, Argonne National Laboratory, 12 May 2026 (virtual).
- "Efficient Machine Learning for Physics: Solving Inverse Problems and Identifying Particles", HEP Seminar, Department of Physics, Indian Institute of Technology Kanpur, 9 December 2025.

- “Bayesian Model Comparison: The Battle Royale of New Physics Models”, PHOENIX 2025, Indian Institute of Technology Hyderabad, 9–11 July 2025.
- “Optimizing Probes of New Physics Models: Generative Algorithm-Driven Nested Sampling for Parameter Estimation”, From Big Bang to Now: A Theory-Experiment Dialogue, SRM University AP, 23–25 January 2025.
- “Boosting Beyond Standard Model Searches: ML-Enhanced Nested Sampling for Rapid Parameter Estimation”, XXVI DAE-BRNS High Energy Physics Symposium, Banaras Hindu University, 19–23 December 2024.
- “Seeking Higgsino-like Neutralino guided by GNN-based Top Tagger”, XXVI DAE-BRNS High Energy Physics Symposium, Banaras Hindu University, 19–23 December 2024.

Selected Conferences, Workshops & Schools

- *PHOENIX 2025*, IIT Hyderabad, 9–11 July 2025.
- *Current Trends in Particle and Astroparticle Physics*, IACS Kolkata, 30–31 January 2025.
- *From Big Bang to Now: A Theory-Experiment Dialogue*, SRM University AP, 23–25 January 2025.
- *XXVI DAE-BRNS High Energy Physics Symposium*, BHU Varanasi, 19–23 December 2024.
- *Machine Learning for Particle and Astroparticle Physics*, IoP Bhubaneswar, 1–13 July 2024.
- *Workshop on Dark Matter and Astroparticle Physics*, IoP Bhubaneswar, 7–9 August 2024.
- *SANGAM@HRI 2024*, HRI Prayagraj, 7–16 March 2024.
- *Statistical Methods and Machine Learning in High Energy Physics*, ICTS-TIFR, 28 August–8 September 2023.
- *SAGEX School in Amplitudes*, 9–13 August 2021.
- *Theoretical Physics School @ IIT Gandhinagar*, 7–11 June 2021.

Teaching

- Tutorial instructor: Normalizing Flows and Simulation-Based Inference, ML4HEP V3 2025, IISER Kolkata.
- Undergraduate tutorials: Newtonian Mechanics and Electrodynamics.
- Undergraduate laboratory instruction: Solid State, Optics, Mechanics, and Electrodynamics.

References

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